Definitive Comparison of Every Classifier

Same three-seed ensemble calibration rule applied to every valid frozen configuration

WINNER

Weighted HYB-MS +10% MS-HardGAN-C

Evaluation	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
Frozen 10K	93.10%	98.43%	87.60%	92.70%	96.89%	97.81%	1.40%	12.40%
Imbalanced 30K	95.43%	97.49%	87.05%	91.98%	96.99%	96.38%	0.96%	12.95%

- Ranked first in F1, accuracy, ROC-AUC, and PR-AUC on both external sets after every model received the same calibration rule.
- Its FPR remains below the declared 2% deployment-candidate screen on both sets: 1.40% on 10K and 0.96% on 30K.
- It beats weighted real-only HYB-MS, ordinary MS-LSTM, natural HYB-MS, every CNN/RNN/RF/SVM condition, and every higher GAN dose.
- This is a retrospective final selection. A new untouched temporal snapshot is required to confirm the selected winner.

Controlled Final Protocol

The threshold-policy confound has now been removed

Fixed element	Value	Why it matters
Configurations	106	Every valid three-seed classifier condition
Checkpoints	318	Seeds 42/43/44 ensembled by mean score
Calibration	Same separate 400 URLs	No hybrid-only threshold advantage
Threshold rule	Max F1 subject to calibration FPR ≤1%	Identical rule for all score scales
Internal test	49,229 frozen URLs	Familiar-distribution reference
External 10K	10,000 frozen URLs	Balanced source-transfer evaluation
External 30K	30,000 URLs; 30.07% phishing	Later-collected imbalanced stress test

What changed

Previously the hybrid used cross-source validation/calibration while ordinary baselines retained their original internal thresholds. This definitive run recalibrated every three-seed ensemble on the exact same 400-row set.

What did not change

No model was retrained, no checkpoint was selected from external labels, and no external threshold was tuned. Architectures, features, source exposure, augmentation pools, and learned weights remain the actual experimental conditions being compared.

Statistical guardrail: calibration resubstitution is not reported as test performance. The selected winner must be evaluated once more on a future untouched cohort because the 10K/30K are now used for retrospective model selection.

Top Models — Frozen External 10K

Identical calibration; sorted by F1

#	Stage	Model	Condition	Acc.	Prec.	Recall	F1	AUC	FPR
1	Feature-equal / weighted HYB	HYB-MS	+10%	93.10%	98.43%	87.60%	92.70%	96.89%	1.40%
2	Weighted HYB-MS	HYB-MS	Real	91.77%	98.04%	85.24%	91.20%	96.63%	1.70%
3	Feature-equal / weighted HYB	HYB-MS	+20%	91.30%	98.96%	83.48%	90.56%	96.30%	0.88%
4	Ordinary MS	LSTM	Real	90.63%	98.54%	82.48%	89.80%	93.72%	1.22%
5	Feature-equal / weighted HYB	HYB-MS	+30%	88.39%	99.23%	77.38%	86.95%	95.85%	0.60%
6	MS-HardGAN-C baselines	LSTM	+40%	85.16%	98.14%	71.68%	82.85%	91.45%	1.36%
7	Feature-equal / weighted HYB	HYB-MS	+40%	85.15%	99.30%	70.80%	82.66%	95.55%	0.50%
8	Feature-equal / weighted HYB	RF	Real	84.45%	96.39%	71.58%	82.15%	91.12%	2.68%
9	MS-HardGAN-C baselines	LSTM	+20%	83.33%	97.76%	68.22%	80.36%	90.58%	1.56%
10	MS-HardGAN-C baselines	LSTM	+10%	82.62%	97.58%	66.90%	79.38%	89.95%	1.66%
11	Coverage	LSTM	Real + outside legit	81.94%	97.19%	65.78%	78.46%	91.23%	1.90%
12	Natural HYB-MS	HYB-MS	Real	81.42%	98.37%	63.90%	77.47%	94.22%	1.06%
13	Feature-equal / weighted HYB	CNN+PSL	+10%	80.58%	96.93%	63.16%	76.48%	91.96%	2.00%
14	MS-HardGAN-C baselines	LSTM	+30%	80.08%	99.05%	60.74%	75.30%	91.19%	0.58%
15	Feature-equal / weighted HYB	CNN+PSL	Real	79.36%	97.76%	60.10%	74.44%	93.29%	1.38%

F1 is the primary balance metric here. FPR remains a separate deployment constraint; a model is not selected merely because it minimizes false alarms by refusing to detect phishing.

Top Models — Imbalanced External 30K

Identical calibration; sorted by F1

#	Stage	Model	Condition	Acc.	Prec.	Recall	F1	AUC	FPR
1	Feature-equal / weighted HYB	HYB-MS	+10%	95.43%	97.49%	87.05%	91.98%	96.99%	0.96%
2	Weighted HYB-MS	HYB-MS	Real	94.64%	96.51%	85.25%	90.53%	96.94%	1.33%
3	Feature-equal / weighted HYB	HYB-MS	+20%	94.23%	98.27%	82.26%	89.56%	96.58%	0.62%
4	Ordinary MS	LSTM	Real	93.96%	96.31%	83.11%	89.22%	93.95%	1.37%
5	Feature-equal / weighted HYB	HYB-MS	+30%	92.07%	98.47%	74.78%	85.01%	95.92%	0.50%
6	MS-HardGAN-C baselines	LSTM	+40%	91.90%	97.01%	75.40%	84.85%	91.94%	1.00%
7	Coverage	LSTM	Real + outside legit	91.62%	94.53%	76.58%	84.61%	92.13%	1.91%
8	MS-HardGAN-C baselines	LSTM	+20%	91.34%	96.33%	74.01%	83.71%	91.66%	1.21%
9	MS-HardGAN-C baselines	LSTM	+10%	91.23%	95.67%	74.20%	83.58%	90.83%	1.44%
10	Feature-equal / weighted HYB	HYB-MS	+40%	90.85%	98.46%	70.69%	82.29%	95.71%	0.48%
11	MS-HardGAN-C baselines	LSTM	+30%	90.51%	98.29%	69.65%	81.53%	91.76%	0.52%
12	Natural HYB-MS	HYB-MS	Real	89.15%	97.22%	65.81%	78.49%	94.70%	0.81%
13	Feature-equal / weighted HYB	RF	Real	88.43%	90.74%	68.53%	78.09%	89.78%	3.01%
14	Natural HYB-MS	HYB-MS	+10%	88.04%	95.01%	63.57%	76.18%	92.87%	1.43%
15	SeqGAN v1	LSTM	Coverage +10%	87.59%	96.09%	61.22%	74.79%	86.72%	1.07%

F1 is the primary balance metric here. FPR remains a separate deployment constraint; a model is not selected merely because it minimizes false alarms by refusing to detect phishing.

Winners by Experimental Category

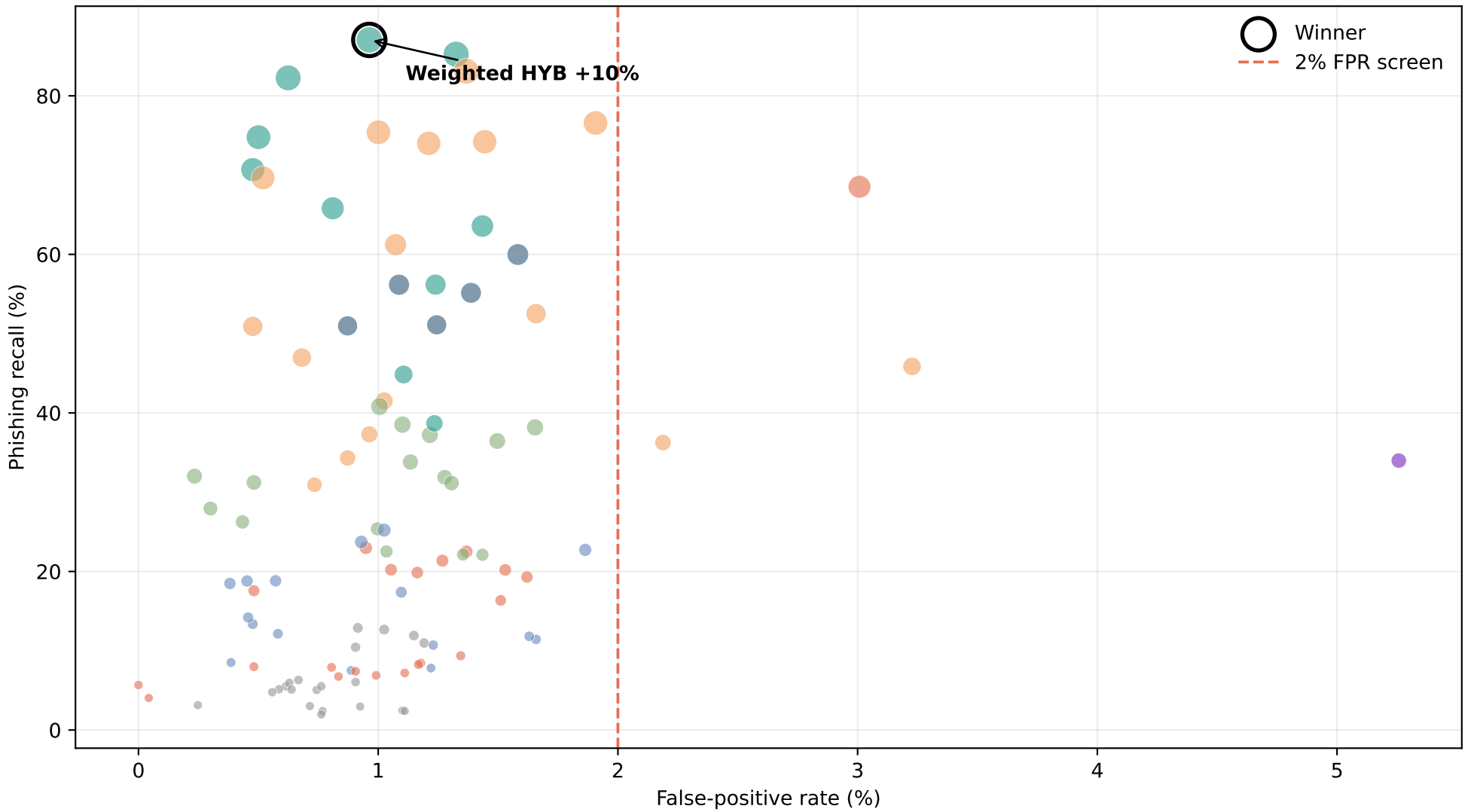
FPR ≤2% on both external sets; highest mean external F1 within category

Category	Stage	Model	Condition	10K F1	30K F1	10K FPR	30K FPR
Best overall	Feature-equal / weighted HYB	HYB-MS	+10%	92.70%	91.98%	1.40%	0.96%
Best without GAN	Weighted HYB-MS	HYB-MS	Real	91.20%	90.53%	1.70%	1.33%
Best non-hybrid	Ordinary MS	LSTM	Real	89.80%	89.22%	1.22%	1.37%
Best natural HYB	Natural HYB-MS	HYB-MS	Real	77.47%	78.49%	1.06%	0.81%
Best CNN+PSL	Feature-equal / weighted HYB	CNN+PSL	+10%	76.48%	73.30%	2.00%	1.58%
Best RF	Feature-equal / weighted HYB	RF	+10%	47.98%	36.71%	1.02%	0.95%
Best original SeqGAN	SeqGAN v1	LSTM	Coverage +10%	68.07%	74.79%	1.22%	1.07%
Best MS-HardGAN baseline	MS-HardGAN-C baselines	LSTM	+40%	82.85%	84.85%	1.36%	1.00%

Best non-hybrid: ordinary MS-LSTM. Best natural hybrid: real-only Natural HYB-MS. Best no-GAN complete system: weighted real-only HYB-MS. None exceeds weighted HYB-MS +10%.

Deployment Trade-off on the 30K

Recall versus false-positive rate; marker size represents F1



The winner occupies the desirable upper-left region: high recall with low FPR. Higher weighted GAN doses move left but also sharply downward, while many simple models remain low-recall after equal calibration.

Finalists: Direct Head-to-Head

The strongest system from each major design family

Finalist	10 F1	10 R	10 FPR	10 AUC	30 F1	30 R	30 FPR	30 AUC
Feature-equal / weighted HYB / HYB-MS +10%	92.70%	87.60%	1.40%	96.89%	91.98%	87.05%	0.96%	96.99%
Weighted HYB-MS / HYB-MS Real	91.20%	85.24%	1.70%	96.63%	90.53%	85.25%	1.33%	96.94%
Ordinary MS / LSTM Real	89.80%	82.48%	1.22%	93.72%	89.22%	83.11%	1.37%	93.95%
Natural HYB-MS / HYB-MS Real	77.47%	63.90%	1.06%	94.22%	78.49%	65.81%	0.81%	94.70%
Feature-equal / weighted HYB / CNN+PSL +10%	76.48%	63.16%	2.00%	91.96%	73.30%	59.98%	1.58%	91.75%
Feature-equal / weighted HYB / RF +10%	47.98%	31.88%	1.02%	58.77%	36.71%	22.98%	0.95%	52.12%

Why +10% rather than +20/+30/+40%?

The larger doses achieve lower FPR and therefore can rank well under equal-weight metric averages, but they miss substantially more phishing. +10% gives the highest F1, recall, accuracy, ROC-AUC, and PR-AUC on both external sets while keeping FPR below 2%.

Selection principle: maximize useful phishing detection under a practical false-alarm ceiling—not maximize one metric in isolation.

Internal Success Does Not Guarantee External Success

Top internal F1 configurations compared with their 30K F1

#	Stage	Model	Condition	Internal F1	Internal AUC	30K F1	30K AUC	30K FPR
1	Coverage	LSTM	Real + outside legit	94.15%	99.80%	84.61%	92.13%	1.91%
2	Ordinary MS	LSTM	Real	93.11%	99.66%	89.22%	93.95%	1.37%
3	SeqGAN v1	LSTM	Coverage +10%	92.83%	99.59%	74.79%	86.72%	1.07%
4	MS-HardGAN-C baselines	LSTM	+40%	91.92%	99.69%	84.85%	91.94%	1.00%
5	MS-HardGAN-C baselines	LSTM	+10%	91.72%	99.57%	83.58%	90.83%	1.44%
6	MS-HardGAN-C baselines	LSTM	+20%	91.71%	99.67%	83.71%	91.66%	1.21%
7	Natural HYB-MS	HYB-MS	+10%	91.34%	99.74%	76.18%	92.87%	1.43%
8	SeqGAN v1	LSTM	Coverage +40%	91.05%	99.60%	59.81%	74.22%	3.23%
9	SeqGAN v1	LSTM	Coverage +30%	91.05%	99.70%	67.17%	80.64%	1.66%
10	Natural HYB-MS	HYB-MS	Real	90.91%	99.83%	78.49%	94.70%	0.81%
11	MS-HardGAN-C baselines	LSTM	+30%	90.82%	99.65%	81.53%	91.76%	0.52%
12	Feature-equal / weighted HYB	HYB-MS	+10%	90.82%	99.55%	91.98%	96.99%	0.96%
13	SeqGAN v1	LSTM	Coverage +20%	90.29%	99.67%	66.98%	82.88%	0.48%
14	Natural HYB-MS	HYB-MS	+20%	90.14%	99.74%	70.64%	91.28%	1.24%
15	Feature-equal / weighted HYB	CNN+PSL	+10%	90.05%	99.76%	73.30%	91.75%	1.58%

This page explains why the project moved beyond internal accuracy: several near-perfect internal models transfer poorly. The final winner is chosen from external behavior after equal calibration.

Pros and Cons of the Main Model Classes

A concise interpretation of all 106 configurations

Class	Pros	Cons	Verdict
CNN	Fast local-pattern learner	Calibration often suppresses recall; weak source transfer	Baseline/ablation
LSTM	Best non-hybrid sequence family; MS real reaches 89.22% 30K F1	Recurrent latency; still below weighted hybrid	Strong fallback
RNN	Simple recurrent control	Unstable and consistently behind LSTM	Research control
RF	Interpretable lexical baseline	GAN doses and source shift hurt ranking	Explainability baseline
SVM	Small, reproducible linear model	Lowest external discrimination in most conditions	Lower bound
CNN+PSL	Tests character/lexical fusion without full adaptation	External ceiling below HYB-MS	Feature-fusion ablation
Natural HYB-MS	Fair row frequency; low FPR; strong AUC	Less outside exposure; 30K F1 78.49%	Natural-sampling winner
Weighted HYB real	Excellent calibrated transfer without synthetic data	Repeats 5,073 outside URLs heavily	Best no-GAN system
Weighted HYB +10%	Best balanced evidence on both external sets	Most complex; synthetic dependency	Overall winner
Weighted HYB 20–40%	Very low FPR	Increasing FNR and decreasing F1	Not preferred

Generator-only SeqGAN, URL-GAN, RelGAN, and MS-HardGAN-C are not classifiers and therefore are not assigned classifier Accuracy/F1. Their role is evaluated through acceptance, uniqueness, realism, coverage, and challenge metrics.

Confidence and Claim Guardrails

What the definitive leaderboard supports—and what it does not

Set	Recall	95% Wilson CI	FPR	95% Wilson CI	Counts
10K	87.60%	86.66%–88.48%	1.40%	1.11%–1.76%	TP 4380; FP 70
30K	87.05%	86.34%–87.73%	0.96%	0.84%–1.10%	TP 7853; FP 202

SUPPORTED

- Under one identical calibration protocol, weighted HYB-MS +10% is the best balanced classifier tested.
- Its advantage appears on two identity-disjoint external datasets with different class prevalence.
- Weighted exposure, CORAL, feature fusion, ensembling, calibration, and low-dose GAN augmentation jointly define the winning system.

NOT YET SUPPORTED

- A claim that the hybrid architecture alone caused the result.
- A claim that GANs predict fundamentally unseen zero-day concepts.
- A final unbiased deployment claim after using the 10K/30K to choose the winner; a new temporal holdout is required.

Appendix: Every Classifier Configuration (1/6)

All rows use the same three-seed calibration rule

Stage	Model	Cond.	10 F1	10 R	10 FP	10 AUC	30 F1	30 R	30 FP	30 AUC
Coverage	CNN	Real + outside legit	26.85%	15.78%	1.76%	63.24%	35.77%	22.72%	1.86%	70.83%
Coverage	LSTM	Real + outside legit	78.46%	65.78%	1.90%	91.23%	84.61%	76.58%	1.91%	92.13%
Coverage	RF	Real + outside legit	20.24%	11.44%	1.60%	46.06%	16.65%	9.37%	1.34%	39.67%
Coverage	RNN	Real + outside legit	43.99%	28.52%	1.16%	82.37%	49.56%	33.81%	1.13%	82.15%
Coverage	SVM	Real + outside legit	13.05%	7.06%	1.14%	24.02%	20.80%	11.92%	1.15%	30.07%
Feature-equal / weighted HYB	CNN+PSL	+10%	76.48%	63.16%	2.00%	91.96%	73.30%	59.98%	1.58%	91.75%
Feature-equal / weighted HYB	CNN+PSL	+20%	71.21%	56.22%	1.68%	90.71%	69.64%	55.15%	1.39%	91.05%
Feature-equal / weighted HYB	CNN+PSL	+30%	67.56%	51.62%	1.20%	91.63%	66.64%	50.98%	0.87%	91.56%
Feature-equal / weighted HYB	CNN+PSL	+40%	66.79%	50.96%	1.64%	89.83%	66.38%	51.11%	1.24%	90.27%
Feature-equal / weighted HYB	CNN+PSL	Real	74.44%	60.10%	1.38%	93.29%	70.79%	56.17%	1.09%	93.22%
Feature-equal / weighted HYB	HYB-MS	+10%	92.70%	87.60%	1.40%	96.89%	91.98%	87.05%	0.96%	96.99%
Feature-equal / weighted HYB	HYB-MS	+20%	90.56%	83.48%	0.88%	96.30%	89.56%	82.26%	0.62%	96.58%
Feature-equal / weighted HYB	HYB-MS	+30%	86.95%	77.38%	0.60%	95.85%	85.01%	74.78%	0.50%	95.92%
Feature-equal / weighted HYB	HYB-MS	+40%	82.66%	70.80%	0.50%	95.55%	82.29%	70.69%	0.48%	95.71%
Feature-equal / weighted HYB	RF	+10%	47.98%	31.88%	1.02%	58.77%	36.71%	22.98%	0.95%	52.12%
Feature-equal / weighted HYB	RF	+20%	35.60%	22.02%	1.68%	51.90%	27.28%	16.35%	1.51%	45.27%
Feature-equal / weighted HYB	RF	+30%	40.46%	25.42%	0.24%	54.96%	29.61%	17.57%	0.48%	46.86%
Feature-equal / weighted HYB	RF	+40%	44.86%	29.34%	1.46%	56.02%	35.79%	22.49%	1.37%	46.99%

Appendix: Every Classifier Configuration (2/6)

All rows use the same three-seed calibration rule

Stage	Model	Cond.	10 F1	10 R	10 FP	10 AUC	30 F1	30 R	30 FP	30 AUC
Feature-equal / weighted HYB	RF	Real	82.15%	71.58%	2.68%	91.12%	78.09%	68.53%	3.01%	89.78%
Feature-equal / weighted HYB	SVM	+10%	7.57%	3.98%	1.20%	55.12%	11.19%	6.05%	0.91%	43.08%
Feature-equal / weighted HYB	SVM	+20%	4.27%	2.20%	0.90%	49.58%	5.78%	3.03%	0.72%	37.44%
Feature-equal / weighted HYB	SVM	+30%	3.89%	2.00%	0.94%	44.64%	4.60%	2.39%	0.77%	33.70%
Feature-equal / weighted HYB	SVM	+40%	4.53%	2.34%	0.98%	40.42%	3.78%	1.96%	0.76%	31.64%
Feature-equal / weighted HYB	SVM	Real	16.51%	9.10%	1.14%	62.97%	18.56%	10.44%	0.91%	52.95%
HYB-ORIG ablation	HYB	Real	47.25%	32.52%	5.12%	73.05%	46.49%	33.99%	5.26%	70.64%
MS-HardGAN-C baselines	CNN	+10%	22.67%	12.88%	0.74%	76.61%	31.33%	18.82%	0.57%	79.23%
MS-HardGAN-C baselines	CNN	+20%	20.55%	11.50%	0.42%	73.37%	30.98%	18.49%	0.38%	77.64%
MS-HardGAN-C baselines	CNN	+30%	31.79%	19.14%	1.28%	76.35%	39.53%	25.22%	1.02%	78.66%
MS-HardGAN-C baselines	CNN	+40%	31.29%	18.76%	1.16%	74.73%	37.69%	23.72%	0.93%	77.95%
MS-HardGAN-C baselines	LSTM	+10%	79.38%	66.90%	1.66%	89.95%	83.58%	74.20%	1.44%	90.83%
MS-HardGAN-C baselines	LSTM	+20%	80.36%	68.22%	1.56%	90.58%	83.71%	74.01%	1.21%	91.66%
MS-HardGAN-C baselines	LSTM	+30%	75.30%	60.74%	0.58%	91.19%	81.53%	69.65%	0.52%	91.76%
MS-HardGAN-C baselines	LSTM	+40%	82.85%	71.68%	1.36%	91.45%	84.85%	75.40%	1.00%	91.94%
MS-HardGAN-C baselines	RF	+10%	49.94%	33.72%	1.32%	57.67%	34.37%	21.36%	1.27%	46.29%
MS-HardGAN-C baselines	RF	+20%	47.53%	31.70%	1.68%	55.96%	32.64%	20.20%	1.53%	44.43%
MS-HardGAN-C baselines	RF	+30%	44.94%	29.48%	1.72%	54.90%	31.36%	19.30%	1.62%	43.53%

Appendix: Every Classifier Configuration (3/6)

All rows use the same three-seed calibration rule

Stage	Model	Cond.	10 F1	10 R	10 FP	10 AUC	30 F1	30 R	30 FP	30 AUC
MS-HardGAN-C baselines	RF	+40%	45.00%	29.42%	1.34%	55.21%	32.41%	19.86%	1.16%	43.67%
MS-HardGAN-C baselines	RNN	+10%	37.75%	23.32%	0.24%	82.90%	43.43%	27.93%	0.30%	85.12%
MS-HardGAN-C baselines	RNN	+20%	44.46%	29.08%	1.72%	79.75%	53.76%	38.18%	1.65%	83.04%
MS-HardGAN-C baselines	RNN	+30%	37.75%	23.38%	0.50%	83.58%	41.25%	26.25%	0.43%	84.25%
MS-HardGAN-C baselines	RNN	+40%	39.39%	24.68%	0.64%	80.61%	47.19%	31.23%	0.48%	83.09%
MS-HardGAN-C baselines	SVM	+10%	3.30%	1.68%	0.26%	17.53%	6.05%	3.14%	0.25%	21.74%
MS-HardGAN-C baselines	SVM	+20%	2.97%	1.52%	0.82%	14.59%	5.63%	2.96%	0.92%	18.00%
MS-HardGAN-C baselines	SVM	+30%	2.20%	1.12%	0.92%	12.92%	4.69%	2.46%	1.10%	15.77%
MS-HardGAN-C baselines	SVM	+40%	2.08%	1.06%	0.94%	12.52%	4.56%	2.39%	1.11%	15.28%
Natural HYB-MS	HYB-MS	+10%	74.16%	60.08%	1.94%	91.91%	76.18%	63.57%	1.43%	92.87%
Natural HYB-MS	HYB-MS	+20%	67.10%	51.08%	1.16%	90.36%	70.64%	56.18%	1.24%	91.28%
Natural HYB-MS	HYB-MS	+30%	56.83%	40.16%	1.18%	89.05%	60.85%	44.85%	1.11%	90.21%
Natural HYB-MS	HYB-MS	+40%	50.71%	34.42%	1.32%	86.27%	54.66%	38.69%	1.23%	87.69%
Natural HYB-MS	HYB-MS	Real	77.47%	63.90%	1.06%	94.22%	78.49%	65.81%	0.81%	94.70%
ORIG baseline	CNN	Real	13.31%	7.24%	1.58%	56.18%	19.83%	11.43%	1.66%	62.44%
ORIG baseline	LSTM	Real	55.24%	38.62%	1.20%	80.46%	57.70%	41.51%	1.02%	84.00%
ORIG baseline	RF	Real	8.75%	4.58%	0.06%	24.76%	7.77%	4.05%	0.04%	21.57%
ORIG baseline	RNN	Real	48.12%	32.04%	1.12%	79.47%	47.30%	31.89%	1.28%	76.02%

Appendix: Every Classifier Configuration (4/6)

All rows use the same three-seed calibration rule

Stage	Model	Cond.	10 F1	10 R	10 FP	10 AUC	30 F1	30 R	30 FP	30 AUC
ORIG baseline	SVM	Real	13.05%	7.06%	1.10%	23.42%	19.30%	10.97%	1.19%	29.46%
Ordinary MS	CNN	Real	20.50%	11.48%	0.54%	79.83%	31.37%	18.80%	0.45%	82.29%
Ordinary MS	LSTM	Real	89.80%	82.48%	1.22%	93.72%	89.22%	83.11%	1.37%	93.95%
Ordinary MS	RF	Real	47.51%	31.56%	1.30%	63.01%	32.97%	20.22%	1.05%	53.12%
Ordinary MS	RNN	Real	51.15%	34.78%	1.20%	89.01%	57.02%	40.82%	1.01%	88.27%
Ordinary MS	SVM	Real	12.89%	6.96%	1.06%	24.99%	22.03%	12.67%	1.02%	30.92%
SeqGAN v1	CNN	Coverage +10%	18.36%	10.22%	1.12%	57.38%	28.99%	17.38%	1.10%	63.72%
SeqGAN v1	CNN	Coverage +20%	13.65%	7.36%	0.44%	54.09%	21.39%	12.14%	0.58%	59.30%
SeqGAN v1	CNN	Coverage +30%	15.75%	8.58%	0.40%	55.06%	23.36%	13.37%	0.48%	59.74%
SeqGAN v1	CNN	Coverage +40%	17.00%	9.34%	0.56%	51.21%	24.64%	14.20%	0.46%	57.00%
SeqGAN v1	CNN	ORIG +10%	14.79%	8.10%	1.46%	53.96%	20.44%	11.82%	1.63%	59.81%
SeqGAN v1	CNN	ORIG +20%	13.55%	7.36%	1.30%	49.83%	18.88%	10.72%	1.23%	57.23%
SeqGAN v1	CNN	ORIG +30%	10.44%	5.58%	1.28%	45.94%	14.13%	7.82%	1.22%	53.66%
SeqGAN v1	CNN	ORIG +40%	9.94%	5.28%	0.94%	46.48%	13.72%	7.52%	0.89%	54.16%
SeqGAN v1	LSTM	Coverage +10%	68.07%	52.22%	1.22%	78.89%	74.79%	61.22%	1.07%	86.72%
SeqGAN v1	LSTM	Coverage +20%	60.02%	43.08%	0.48%	77.33%	66.98%	50.91%	0.48%	82.88%
SeqGAN v1	LSTM	Coverage +30%	59.12%	42.64%	1.60%	73.85%	67.17%	52.52%	1.66%	80.64%
SeqGAN v1	LSTM	Coverage +40%	53.43%	37.64%	3.26%	65.16%	59.81%	45.87%	3.23%	74.22%

Appendix: Every Classifier Configuration (5/6)

All rows use the same three-seed calibration rule

Stage	Model	Cond.	10 F1	10 R	10 FP	10 AUC	30 F1	30 R	30 FP	30 AUC
SeqGAN v1	LSTM	ORIG +10%	52.41%	35.84%	0.92%	78.51%	53.46%	37.30%	0.96%	76.51%
SeqGAN v1	LSTM	ORIG +20%	48.88%	32.58%	0.72%	65.18%	50.34%	34.32%	0.87%	67.89%
SeqGAN v1	LSTM	ORIG +30%	48.56%	32.74%	2.10%	65.94%	51.31%	36.26%	2.19%	66.07%
SeqGAN v1	LSTM	ORIG +40%	39.22%	24.58%	0.78%	65.46%	46.65%	30.94%	0.73%	68.84%
SeqGAN v1	RF	Coverage +10%	18.06%	10.06%	1.32%	36.77%	15.19%	8.45%	1.18%	32.74%
SeqGAN v1	RF	Coverage +20%	16.26%	8.94%	1.02%	32.38%	14.40%	7.90%	0.81%	29.37%
SeqGAN v1	RF	Coverage +30%	16.62%	9.18%	1.30%	30.48%	14.90%	8.27%	1.17%	28.03%
SeqGAN v1	RF	Coverage +40%	15.89%	8.68%	0.58%	29.86%	14.65%	7.99%	0.48%	27.72%
SeqGAN v1	RF	ORIG +10%	13.58%	7.38%	1.28%	21.11%	12.63%	6.90%	0.99%	18.98%
SeqGAN v1	RF	ORIG +20%	13.07%	7.06%	1.00%	19.84%	12.42%	6.75%	0.83%	18.21%
SeqGAN v1	RF	ORIG +30%	11.21%	5.94%	0.00%	19.47%	10.74%	5.68%	0.00%	18.01%
SeqGAN v1	RF	ORIG +40%	14.29%	7.78%	1.14%	19.39%	13.52%	7.40%	0.91%	18.00%
SeqGAN v1	RNN	Coverage +10%	48.75%	32.76%	1.64%	82.62%	53.15%	37.21%	1.22%	82.04%
SeqGAN v1	RNN	Coverage +20%	45.51%	29.96%	1.70%	74.45%	52.11%	36.46%	1.50%	79.07%
SeqGAN v1	RNN	Coverage +30%	57.82%	41.16%	1.22%	80.93%	54.61%	38.52%	1.10%	80.78%
SeqGAN v1	RNN	Coverage +40%	36.80%	22.62%	0.30%	76.97%	48.32%	32.03%	0.23%	78.13%
SeqGAN v1	RNN	ORIG +10%	53.56%	36.92%	0.94%	80.49%	36.06%	22.53%	1.03%	76.92%
SeqGAN v1	RNN	ORIG +20%	42.35%	27.22%	1.32%	72.37%	35.24%	22.10%	1.43%	71.61%

Appendix: Every Classifier Configuration (6/6)

All rows use the same three-seed calibration rule

Stage	Model	Cond.	10 F1	10 R	10 FP	10 AUC	30 F1	30 R	30 FP	30 AUC
SeqGAN v1	RNN	ORIG +30%	54.48%	37.78%	0.92%	79.74%	39.74%	25.37%	1.00%	78.03%
SeqGAN v1	RNN	ORIG +40%	55.19%	38.60%	1.28%	74.25%	46.43%	31.15%	1.31%	77.33%
SeqGAN v1	SVM	Coverage +10%	8.20%	4.30%	0.64%	19.69%	11.70%	6.31%	0.67%	24.58%
SeqGAN v1	SVM	Coverage +20%	6.83%	3.56%	0.62%	17.03%	10.30%	5.51%	0.61%	20.84%
SeqGAN v1	SVM	Coverage +30%	6.46%	3.36%	0.60%	15.16%	9.66%	5.14%	0.59%	18.20%
SeqGAN v1	SVM	Coverage +40%	6.50%	3.38%	0.70%	13.86%	9.47%	5.05%	0.74%	16.51%
SeqGAN v1	SVM	ORIG +10%	7.87%	4.12%	0.60%	19.18%	11.06%	5.94%	0.63%	23.79%
SeqGAN v1	SVM	ORIG +20%	6.98%	3.64%	0.70%	16.69%	10.29%	5.52%	0.76%	20.20%
SeqGAN v1	SVM	ORIG +30%	6.43%	3.34%	0.58%	14.91%	9.59%	5.11%	0.64%	17.72%
SeqGAN v1	SVM	ORIG +40%	6.06%	3.14%	0.50%	13.65%	8.99%	4.77%	0.56%	16.17%
Structure-balanced	CNN	Real	10.17%	5.38%	0.42%	61.21%	15.56%	8.51%	0.39%	67.56%
Structure-balanced	LSTM	Real	63.77%	47.24%	0.92%	84.22%	63.24%	46.98%	0.68%	84.16%
Structure-balanced	RF	Real	14.34%	7.82%	1.24%	29.22%	13.11%	7.19%	1.11%	25.11%
Structure-balanced	RNN	Real	34.97%	21.42%	1.08%	76.65%	35.34%	22.14%	1.35%	74.59%
Structure-balanced	SVM	Real	14.80%	8.08%	1.08%	27.76%	22.40%	12.88%	0.92%	33.15%
Weighted HYB-MS	HYB-MS	Real	91.20%	85.24%	1.70%	96.63%	90.53%	85.25%	1.33%	96.94%

Final Decision

One winner, with an explicit deployment boundary

Best model: calibrated three-seed Weighted HYB-MS +10% MS-HardGAN-C ensemble.

Runner-up category	Best system	10K F1	30K F1	Reason it loses
No GAN	Weighted HYB-MS | HYB-MS | Real	91.20%	90.53%	Lower recall/F1 on both sets
Non-hybrid	Ordinary MS | LSTM | Real	89.80%	89.22%	Lower AUC and F1
Natural hybrid	Natural HYB-MS | HYB-MS | Real	77.47%	78.49%	Less source exposure; lower recall

NEXT REQUIRED ACTION

Freeze a genuinely later-created phishing cohort and later legitimate snapshot, then evaluate this already-selected ensemble and threshold exactly once. Do not compare or retune the other 105 configurations on that final confirmation set.

- If confirmed, benchmark browser latency, memory, package size, and alert burden at realistic prevalence.
- If it fails, investigate drift and retrain under a new declared protocol rather than tuning on the failed holdout.
- Report it now as the strongest offline deployment candidate—not as production-ready.